SAHAYAK: Publish a Tested New Sign

Simple guide: from your computer to the live website

Use this guide after a new sign has been collected, trained and tested on your computer. It explains how to put the approved model on the live server and check that the website uses it.

The whole process

1. Keep a backup of the old model.
2. Check the model and its sign names.
3. Copy only the approved model file.
4. Push the change to GitHub.
5. Wait for Render to finish.
6. Check the live model status.
7. Test old and new signs in Live Studio.

There are two separate models

- Static model: recognizes a pose held still. File: model\sahayak_twohand.joblib. It expects 126 numbers.
- Continuous model: recognizes a movement over time. File: model\sahayak_sequence.joblib. It expects 1,320 numbers.

Never copy one model over the other.

1. Set the folder paths

Open VS Code, then Terminal > New Terminal. Use PowerShell. Change RepoRoot to the folder where you cloned SAHAYAK-ISL. Do not run this block until both folders are correct.

~~~powershell
$ErrorActionPreference = "Stop"
$TrainingRoot = "C:\Users\user\3D Objects\New folder\SAHAYAK"
$RepoRoot = "C:\PATH\TO\YOUR\SAHAYAK-ISL"
$TrainingPython = Join-Path $TrainingRoot ".venv\Scripts\python.exe"
if (-not (Test-Path $TrainingPython)) { throw "Python environment not found. Check TrainingRoot." }
if (-not (Test-Path (Join-Path $RepoRoot ".git"))) { throw "GitHub clone not found. Check RepoRoot." }
Set-Location $RepoRoot
$Remote = (git remote get-url origin).Trim()
if ($Remote -notmatch 'github\.com[:/]Mrjohanjyotikuli/SAHAYAK-ISL(\.git)?$') {
throw "This is not the expected SAHAYAK-ISL repository."
}
if (@(git status --porcelain).Count -gt 0) {
throw "There are local changes. Review and save them before continuing."
}
git pull --ff-only origin main
if ($LASTEXITCODE -ne 0) { throw "Could not update main." }
Write-Host "Folders and GitHub remote checked."

If a check fails, stop and correct the path. Do not delete other work to make the command pass.

## 2. Check the model and label names
Run this block to check the active files that exist on your computer. The feature count must match the model type. Check that the new sign is present and old signs are still listed.

~~~powershell
$TrainingRoot = "C:\Users\user\3D Objects\New folder\SAHAYAK"
$TrainingPython = Join-Path $TrainingRoot ".venv\Scripts\python.exe"
@'
import joblib
from pathlib import Path
root = Path(r"C:\Users\user\3D Objects\New folder\SAHAYAK")
checks = [
```
```

```
("STATIC", root / "model" / "sahayak_twohand.joblib", 126),
("CONTINUOUS", root / "model" / "sahayak_sequence.joblib", 1320),
]
for name, path, expected in checks:
if not path.is_file():
print(name, "model not found:", path)
continue
model = joblib.load(path)
features = getattr(model, "n_features_in_", None)
labels = sorted(str(x) for x in model.classes_)
print("\n", name, path.name)
print("Features:", features)
print("Labels:", labels)
if features != expected:
raise SystemExit("Wrong feature count. Do not publish this model.")
'@ | & $TrainingPython -
if ($LASTEXITCODE -ne 0) { throw "Model check failed." }
```

If the new sign is missing or the feature count is wrong, stop. Do not publish that model.

## 3. Back up the current deployment models

The deployment copy is inside the GitHub clone. Check that the folder and source models exist before copying.

~~~powershell
$TrainingRoot = "C:\Users\user\3D Objects\New folder\SAHAYAK"
$RepoRoot = "C:\PATH\TO\YOUR\SAHAYAK-ISL"
$SourceDir = Join-Path $TrainingRoot "model"
$TargetDir = Join-Path $RepoRoot "SAHAYAK-Vercel-Deployment\backend\model"
$Names = @("sahayak_twohand.joblib", "sahayak_sequence.joblib")
foreach ($Name in $Names) {
if (-not (Test-Path (Join-Path $SourceDir $Name))) { throw "Missing $Name" }
}
if (-not (Test-Path $TargetDir)) { throw "Deployment model folder missing." }
$Stamp = Get-Date -Format "yyyyMMdd_HHmmss"
$BackupDir = Join-Path $TrainingRoot "backups\deployment_models_$Stamp"
New-Item -ItemType Directory -Force $BackupDir | Out-Null
foreach ($Name in $Names) {
$Target = Join-Path $TargetDir $Name
if (Test-Path $Target) { Copy-Item $Target $BackupDir -Force }
}
Write-Host "Backup saved to $BackupDir"

```
## 4. Copy only the model you changed
For a static-sign change, copy sahayak_twohand.joblib. For a moving-sign change, copy sahayak_sequence.joblib.
    Leave the other model alone. Never copy a candidate file.
```

~~~powershell
```
$Name = "sahayak_twohand.joblib"
# Use "sahayak_sequence.joblib" for a continuous-only update.
$Source = Join-Path $SourceDir $Name
$Target = Join-Path $TargetDir $Name
Copy-Item $Source $Target -Force
$A = (Get-FileHash $Source -Algorithm SHA256).Hash
$B = (Get-FileHash $Target -Algorithm SHA256).Hash
if ($A -ne $B) { throw "The files do not match. Stop." }
Write-Host "Verified copy: $Name"
```

Do not publish raw CSV files, continuous_gesture_db.db, raw samples, or candidate models. The public website needs the approved model, not your private training records.

## 5. Commit only the model file

First inspect the changes. If you see unrelated files, stop and review them. Do not reset your work.

~~~powershell
Set-Location $RepoRoot
if ((git branch --show-current).Trim() -ne "main") { throw "Not on main." }
~~~

if (@(git diff --cached --name-only).Count -gt 0) { throw "Review already staged files first." }
git status --short

```
For a static-only change, stage and push only this file:
```

~~~powershell
git add -- "SAHAYAK-Vercel-Deployment/backend/model/sahayak_twohand.joblib"
git diff --cached --stat
git diff --cached --name-only
git commit -m "Update static sign model"
git push origin main

For a continuous-only change, use this path instead:
SAHAYAK-Vercel-Deployment/backend/model/sahayak_sequence.joblib. Use a message such as Update continuous sign model.

If Git reports a missing author name or email, set them for this repository using your verified GitHub email. Never put a password or access token in a command. If pushing fails, do not force-push.

6. Wait for Render and check the model

Open the Render service page and wait for the newest deployment to say Live:
https://dashboard.render.com/web/srv-dbaof849v7es739am2j0

Then run:
~~~powershell
Invoke-RestMethod "https://sahayak-backend-krh1.onrender.com/api/status" | ConvertTo-Json -Depth 5

```
For a static change, model_ready should be true and single_sign_labels should include the new sign. For a
    continuous change, sequence_model_ready should be true and sequence_sign_labels should include it. Check the
    actual names, not only the totals.
## 7. Test the live website
Open https://sahayak-eight-lac.vercel.app/ and choose Live Studio. Allow camera access, test the new sign in the
    correct mode, and test several old signs too. Check the label, landmarks, box and response. For a moving
    sign, perform the full movement.
## If something goes wrong
- New label missing: check the active model, copied file, commit and API status.
- Model is not ready: wait for Render and review its latest logs.
- Old signs fail: restore the backup and investigate.
- Camera works but prediction is wrong: hand tracking and model prediction are separate steps.
- Git push fails: read the error. Never force-push.
## Final checklist
[ ] Backup saved. [ ] Correct model type selected. [ ] Labels checked. [ ] Only approved model copied. [ ] File
    hashes match. [ ] Only intended model committed. [ ] Render says Live. [ ] API checked. [ ] Old and new
    signs tested.
This guide starts after training, testing and approving the new sign. A model-only update normally does not
    require changes to frontend code.
```